

Jossue Sarango

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EDUCATION

New Jersey Institute of Technology

Bachelor of Science in Mechanical Engineering, Minor in Electrical Engineering

- Affiliations: Management Leadership for Tomorrow | MLT Career Prep

Newark, NJ

Expected: May 2028

EXPERIENCE

Human Friendly Robotics

Mechanical Design Intern – Startup Robotics

Marlboro, NJ

Mar 2026 – Jun 2026

- Owned mechanical design for Tyler, a 50 lb autonomous floor-tiling robot, from CAD through fabrication support, packaging actuators and sensors into a rugged platform.
- Engineered a custom suction end-effector and servo-actuated tilt mechanism for placing tile, LVT, and VCT flooring; collaborated with controls engineers to route wiring and ensure field serviceability.
- Ran clearance and tolerance analysis across the chassis-to-end-effector interface to secure Tyler's $\pm 1\text{mm}$ placement precision, enabling 100 sq ft/hr installation in live deployment at Plaza Las Américas, Puerto Rico.

PSEG

Project Management Intern – Electric Transmission & Distribution

South Plainfield, NJ

Jun 2026 – Aug 2026

- Authored the civil and layout design review (IFR pageturn) for a 138kV ring-bus substation expansion, validating structural steel alignment, H-frame foundation elevations, and disconnect-switch coordination against geotechnical and survey data for construction.
- Audited legacy foundation drawings and ran field walkdowns across a \$60M+, 5-project substation portfolio, matching CAD drawings to as-built conditions and implementing verification test pits to preemptively resolve structural discrepancies.

PSEG

Public Affairs Intern – State Government

Newark, NJ

May 2025 – Aug 2025

- Built and deployed a centralized stakeholder-tracking dashboard consolidating outreach data across 185 municipalities, adopted by New Jersey's largest utility to coordinate infrastructure-permitting efforts for 2.4M electric and 1.9M gas customers.

PROJECTS

Wearable Companion Robot | Animatronics

Prototyping

- Designed a shoulder-mounted head assembly, packaging a 3D-printed thrust-bearing neck, 2-axis eye gimbal, and synced eyelid linkage within a 120mm envelope while holding the clearances needed for 7 servo-driven degrees of freedom to move without binding.
- Consolidated actuation onto a single head-mounted RP2040, eliminating a dedicated PWM driver IC and isolating servo timing from the Jetson's perception workload.

Baja Society of Automotive Engineers | Chassis

FoS, Fabrication

- Prior impact models showed the team's 4130 steel roll cage carried excess-weight tubing in low-stress regions; executed 12+ multi-axis FEA iterations across 15g front, 10g side, and 8g roll-over impact cases to map high-stress node distributions.
- Leveraged stress maps to redesign optimized tubing profiles, cutting vehicle curb weight by 12% while maintaining a $1.5\times$ factor of safety; hand-fabricated and TIG-welded the resulting spaceframe for the final competition build.

Linear Actuator Hand | Prosthetic

3D Printing, PLA

- Designed to give users with limb loss an affordable, reliable alternative to costly commercial prosthetics; built a single-actuator, fishing-line tendon-driven fist mechanism and modeled the palm, fingers, and forearm cuff in SolidWorks.
- Optimized finger joint geometry through prototype testing to fully restore restricted range of motion, removing structural forearm mass to cut component weight and compiling all design iterations into a 27-page technical report.

SHPE NJIT Mobile App | iOS & Android

Agile, CI, Postgres, RLS

- Led a 3-person team to build the first student-built app in NJIT history in 27 days using React Native and Supabase; owned end-to-end brand design, scaling to 250 active users in month one and capturing national recognition from the CEO of SHPE.
- Engineered 6 Serverless Deno Edge Functions and a Cloudflare Worker layout; executed a 7-stage post-launch refactor that automated CI workflows and rendered production 3D assets utilizing Blender.

LEADERSHIP

Society of Hispanic Professional Engineers | SHPE

New Jersey Institute of Technology

Internal Vice President

Sept 2026 – Present

- Secured \$25,025 in FY26-27 operating funding by designing and executing a targeted outreach campaign across university leadership, building strategic partnerships with Deans, Department Chairs, and Executive Cabinets to support 350+ active members.

SKILLS

CAD/Analysis : SolidWorks, Fusion 360, GD&T, Tolerance Stack-Up, FEA

Manufacturing : TIG Welding, Prototyping, SAE Compliance, Fault Tree Analysis

Programming : Python, MATLAB, SQL, JavaScript, Git, Blender

Robotics : ROS 2, NVIDIA Jetson, LiDAR, Servos, Actuators, Microcontrollers

Languages : English, Spanish, Portuguese, Chinese-Mandarin (HSK 2)